Stochastic Calculus and Option Pricing
Lecture 6 Exercise Sheet

6 Week 6 — Densities, Girsanov, and Measure Changes

Throughout, all density processes are assumed adapted to the stated filtration. Whenever a stochastic exponential is used, distinguish the formal drift calculation from the condition that makes it a true martingale.

Part I — What a Density Really Encodes

Exercise 6.1 (Easy to medium — Reversing a change of measure)**.** Let $Z > 0$ satisfy $\mathbb{E}^{\mathbb{P}}[Z] = 1$, and define $\mathrm{d}\mathbb{Q} = Z\,\mathrm{d}\mathbb{P}$.

(a) Prove that

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}} = \frac{1}{Z} \qquad \text{and} \qquad \mathbb{E}^{\mathbb{Q}}\left[\frac{1}{Z}\right] = 1.$$

(b) Show that for every integrable Y,

$$\mathbb{E}^{\mathbb{P}}[Y] = \mathbb{E}^{\mathbb{Q}}\left[\frac{Y}{Z}\right].$$

Explain why the random variable itself has not changed.

(c) Use Jensen's inequality to prove

$$\mathbb{E}^{\mathbb{P}}[\log Z] \leq 0.$$

Assuming the relevant integrability, prove also $\mathbb{E}^{\mathbb{Q}}[\log Z] \geq 0$.

(d) Characterize the equality case in both inequalities. What does it say about the relation between $\mathbb{P}$ and $\mathbb{Q}$?

Exercise 6.2 (Medium — A density process in a two-period tree)**.** Let

$$\Omega = \{UU, UD, DU, DD\},$$

where the first letter is observed at time 1 and the second at time 2. Under $\mathbb{P}$, all four paths have probability $1/4$. Under $\mathbb{Q}$, their probabilities in the displayed order are

$$\frac{3}{8}, \qquad \frac{1}{8}, \qquad \frac{1}{8}, \qquad \frac{3}{8}.$$

Let $\mathcal{F}_1$ reveal only the first letter, and let $Z_2 = \mathrm{d}\mathbb{Q}/\,\mathrm{d}\mathbb{P}$.

(a) Compute Z_2 on every terminal path and verify $\mathbb{E}^{\mathbb{P}}[Z_2] = 1$.

(b) Compute $Z_1 = \mathbb{E}^{\mathbb{P}}[Z_2 \mid \mathcal{F}_1]$. Explain why Z_1 is the density of the time-1 information under $\mathbb{Q}$ relative to $\mathbb{P}$.

(c) Compute under $\mathbb{Q}$ the conditional probability that the second letter is U, first given that the first letter is U, then given that it is D.

(d) Recover both answers from Bayes' formula

$$\mathbb{Q}(A \mid \mathcal{F}_1) = \frac{\mathbb{E}^{\mathbb{P}}[Z_2 \mathbf{1}_A \mid \mathcal{F}_1]}{Z_1}.$$

Does the second step remain independent of the first under $\mathbb{Q}$?

Part II — Tilting Beyond the Classroom Examples

Exercise 6.3 (Medium — Tilting a correlated Gaussian vector)**.** Under $\mathbb{P}$, let (X, Y) be centered jointly Gaussian with

$$\mathrm{Var}(X) = \sigma_X^2, \qquad \mathrm{Var}(Y) = \sigma_Y^2, \qquad \mathrm{Cov}(X, Y) = \rho\sigma_X\sigma_Y.$$

For $\theta \in \mathbb{R}$, define

$$Z = \exp\left(\theta X - \frac{1}{2}\theta^2\sigma_X^2\right), \qquad \mathrm{d}\mathbb{Q} = Z \, \mathrm{d}\mathbb{P}.$$

(a) Verify that $\mathbb{E}^{\mathbb{P}}[Z] = 1$.

(b) Use the joint moment-generating function to compute $\mathbb{E}^{\mathbb{Q}}[e^{uX+vY}]$.

(c) Identify the mean vector and covariance matrix of (X, Y) under $\mathbb{Q}$. Explain why tilting with X also moves the mean of Y when $\rho \neq 0$.

(d) Determine for which ρ the law of Y is unchanged by this measure change.

Exercise 6.4 (Medium — An adapted tilt made safe by stopping)**.** Let

$$\tau_a = \inf\{t \geq 0 : |W_t| \geq a\} \wedge T, \qquad \phi_t = \theta W_t \mathbf{1}_{\{t \leq \tau_a\}},$$

where $a, T > 0$ and $\theta \in \mathbb{R}$. Define

$$Z_t = \exp\left(\int_0^t \phi_s \, \mathrm{d}W_s - \frac{1}{2}\int_0^t \phi_s^2 \, \mathrm{d}s\right).$$

(a) Show that $|\phi_t| \leq |\theta|a$, and verify Novikov's condition. Conclude that Z is a true martingale.

(b) Define $\mathbb{Q}$ by Z_T. Identify the $\mathbb{Q}$-Brownian motion B and write the $\mathbb{Q}$-dynamics of W.

(c) Describe the drift of W before and after τ_a. For $\theta > 0$, does the pre-stopping drift push the process toward or away from zero?

(d) Explain why stopping is doing genuine mathematical work here rather than merely shortening the notation.

Part III — Designing and Testing a Measure Change

Exercise 6.5 (Medium — Changing one mean-reversion model into another)**.** Under $\mathbb{P}$, suppose

$$\mathrm{d}X_t = \kappa(m - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}, \qquad \kappa, \sigma > 0.$$

We would like X to satisfy under $\mathbb{Q}$

$$\mathrm{d}X_t = \widetilde{\kappa}(\widetilde{m} - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \widetilde{\kappa} > 0.$$

(a) Find the candidate process λ_t in $W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \int_0^t \lambda_s\,\mathrm{d}s$.

(b) Write the corresponding candidate density process, with the correct sign.

(c) Explain why the volatility must remain σ. Could an equivalent Girsanov change turn it into a different constant $\widetilde{\sigma} \neq \sigma$?

(d) The candidate λ_t is generally unbounded. Explain why the algebraic drift match does not by itself prove that $\mathbb{Q}$ exists, and describe how localization by $\tau_n = \inf\{t : |X_t| \geq n\} \wedge T$ helps.

Exercise 6.6 (Medium — Can one Brownian shift price two stocks?)**.** Two traded stocks are driven by the same Brownian motion:

$$\frac{\mathrm{d}S_t^1}{S_t^1} = \mu_1\,\mathrm{d}t + \sigma_1\,\mathrm{d}W_t, \qquad \frac{\mathrm{d}S_t^2}{S_t^2} = \mu_2\,\mathrm{d}t + \sigma_2\,\mathrm{d}W_t,$$

where $\sigma_1, \sigma_2 \neq 0$, and the short rate is r.

(a) If $W_t^{\mathbb{Q}} = W_t + \lambda t$, derive the condition on λ that makes each discounted stock driftless.

(b) Show that a common equivalent martingale measure of this form can exist only if

$$\frac{\mu_1 - r}{\sigma_1} = \frac{\mu_2 - r}{\sigma_2}.$$

Interpret this equality economically.

(c) When the equality holds, identify the density process and explain why the Brownian risk premium is unique.

(d) Suppose the equality fails. Find dollar exposures x_1, x_2 in the two stocks such that $x_1\sigma_1 + x_2\sigma_2 = 0$, while $x_1(\mu_1 - r) + x_2(\mu_2 - r) \neq 0$. Explain why this is an instantaneous arbitrage signal and how the sign of the position can be chosen.